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Predictive Modelling of Residential Property Prices for Real Estate Pricing Decisions

Machine Learning | IT3091

B.Sc. (Hons) in Information Technology

Introduction

Real estate agencies handle large volumes of property information, including physical attributes, location details, construction quality, age, and past sale records. By analyzing this information, an agency can better understand what drives property value and can price properties more consistently than by judgement alone.

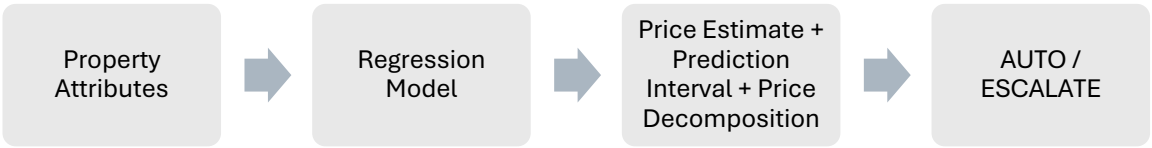
Setting a listing price is one of the most consequential decisions an agency makes. Pricing a property too high leaves it unsold and consumes marketing budget; pricing it too low loses money for the seller and damages the agency's reputation. At present this decision is made either by a professional appraiser, which is accurate but slow and expensive, or by rule of thumb, which is fast but unreliable on unusual properties.

This project applies a machine learning regression technique to estimate the sale price of a residential property from its recorded attributes. Importantly, the project does not stop producing a single predicted number. A single number gives the valuation team no basis for deciding whether to trust it. The output of this project is therefore a short valuation report for each property, consisting of a price estimate, a prediction interval expressing how certain that estimate is, and a breakdown showing which groups of property attributes the value is coming from.

The prediction interval carries the decision the project is designed to support. A narrow interval indicates an ordinary property that the model can price confidently, and such a property can be priced automatically. A wide interval indicates an unusual property whose estimate should not be trusted, and such a property should be routed to a human appraiser. The project therefore helps the agency direct scarce appraisal capacity towards the properties that actually need it.

The project uses the Ames Housing dataset, published on Kaggle as *House Prices: Advanced Regression Techniques*, with the individual property sale taken as the unit of analysis.

The proposed approach is:



The project aims to develop and evaluate a regression model that produces trustworthy, self-assessing price estimates. It will not implement a production pricing system or a deployed application; the outputs serve to provide decision support to the valuation team.

Selected Track

Selected Track: Guided Data Track

Our group selected the Guided Data Track because the module team provided the business domain, the business scenario and the dataset for group code 8, allowing the group to concentrate on the reasoning, modelling and evaluation work rather than on sourcing and validating a dataset.

The Ames Housing dataset is well suited to this project because it records a large number of descriptive attributes per property, covering location, lot characteristics, dwelling type, size, construction quality and condition, age, amenities, and the circumstances of the sale, together with the actual sale price achieved. This makes it possible to investigate not only whether prices can be predicted but also how confidently it can be predicted for different kinds of property.

Our group decided to work with the labelled training file only, rather than attempting to use the supplied unlabeled test file for evaluation. This is explained in the Dataset section and gives a well-defined and honest evaluation scope.

Business Domain

Domain: Real Estate

The selected business domain is Real Estate.

A real estate agency must set a listing price for every property it takes on. This decision is made under time pressure, is difficult to make consistently across different agents, and has a direct financial consequence for both the agency and the seller. Agencies employ or contract professional appraisers, but appraisal capacity is finite and expensive, and it is currently spent on ordinary and unusual properties alike.

The project is therefore concerned with using recorded property data to support pricing decisions in a data-driven way, and with helping the agency decide where its human appraisal effort is best spent.

Business Problem

The main problem addressed by this project is:

How can a real estate agency estimate the sale price of a residential property from its recorded attributes, and know how much confidence to place in each estimate?

The second half of that question is the part that is usually left out. A model that returns a price for every property, with no indication of reliability, forces the agency into one of two poor

choices: trust every estimate, including the ones produced for properties unlike anything the model has seen, or trust none of them, which makes the model pointless.

A regression model that reports uncertainty alongside each estimate allows for a third option. Properties the model can price confidently are priced automatically. Properties the model cannot price confidently are identified as such and sent to an appraiser. The agency gains speed on most of its portfolio without accepting unreliable numbers on the minority.

The additional price decomposition tells the agent where the estimated value comes from, which makes the estimate reviewable by a person rather than something to be accepted or rejected on faith.

Business Lenses

Several business lenses were considered for this project, based on the lenses listed in the assignment descriptor for group code 8.

Primary Lense

Price Prediction — the group's chosen focus

The primary business lens chosen by the group is Price Prediction.

Under this lens, the machine learning model estimates the sale price of an individual property from its recorded attributes. This is the only one of the three listed lenses whose output is a per-property result that enters the agency's workflow directly.

The group has, however, deliberately extended the task formulation beyond a single predicted number. Our output is a price estimate together with a prediction interval and a price decomposition and the interval is used to classify each property as AUTO or ESCALATE.

Reasons for selecting this lens:

- It produces an output the valuation team acts on, for a specific property, on a specific day.
- Attaching a prediction interval converts the estimate into a decision. Without interval, the agency has a number but no basis for trusting it.
- The decomposition makes the estimate reviewable by a non-technical user, which matters for adoption.
- The formulation gives the project a meaningful evaluation question — not only "how accurate is the model" but "does the model know when it is likely to be wrong" — which is a more demanding and more honest standard.

Secondary Lenses

Valuation Feature Analysis

The first possible lens is Valuation Feature Analysis.

Under this approach, the aim would be to identify which property attributes are most strongly associated with higher sale prices across the market as a whole.

Reasons for considering this lens:

- The dataset contains a large and varied set of property attributes, making a feature-importance study straightforward.
- The findings would be useful to the agency's general market understanding.
- The analysis is well supported by established interpretability techniques.

Although this lens was considered, it was not chosen as the primary lens because its output is market-level understanding rather than a per-property result. The valuation team cannot act on it when a specific property arrives on the books. We note, however, that the price decomposition component of our chosen output recovers most of the value of this lens, and does so at the level of the individual property rather than the market average.

Property Segment Analysis

The second possible lens is Property Segment Analysis.

Under this approach, the aim would be to group properties into segments of similar characteristics and to describe each segment.

Reasons for considering this lens:

- The dataset contains varied property types, neighborhoods, and quality levels.
- Segment descriptions would help the agency organize its portfolio.

This lens was not chosen as the primary lens because a grouping of properties is descriptive and does not by itself support a decision. It is, however, useful in a supporting role, and we have adopted it as our secondary lens for that purpose, as described below.

Dataset

The project will use the Ames Housing dataset, published on Kaggle as *House Prices: Advanced Regression Techniques*.

Source: https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques

File	Rows	Columns	Contains the target?
train.csv	1,460	81	Yes — SalePrice
test.csv	1,459	80	No
data_description.txt	—	—	Variable definitions

Verify every figure in this table against the downloaded files. Do not copy row counts from a blog post or another notebook.

The dataset describes residential property sales and contains attributes covering the physical property, its location, its condition, and the circumstances of its sale.

Important variable groups relevant to the proposed task:

Group	Example variables
Target	SalePrice — the recorded sale price
Location	Neighborhood, MSZoning, LotArea, LotFrontage, Condition1, Condition2
Size	GrLivArea, TotalBsmtSF, 1stFlrSF, 2ndFlrSF, GarageArea, bedroom and bathroom counts
Quality and condition	OverallQual, OverallCond, ExterQual, KitchenQual, BsmtQual, HeatingQC
Age	YearBuilt, YearRemodAdd, GarageYrBlt
Amenities	Fireplaces, CentralAir, PoolArea, deck and porch areas
Sale context	SaleType, SaleCondition, MoSold, YrSold

Not all available variables will necessarily be included in the final model. The exact predictor set will be decided after exploratory data analysis, feature analysis, and an assessment of whether each variable would realistically be known at the time a listing price must be set.

Evaluation Constraint

The supplied test.csv file does not contain the SalePrice column, so it cannot be used to measure model performance. All reported results in this project will therefore come from cross-validation on the labelled rows in train.csv.

This is stated explicitly because it shapes the entire validation design. If a submission is made to the public Kaggle leaderboard at any stage, it will be reported as a sanity check only and never presented as our evaluation.

Unit of Analysis

The individual property sale is the unit of analysis.

Each row in the dataset represents one completed residential property sale, and each prediction produced by the model is for one property.

For example, a property's recorded location, size, construction quality, age, and amenities are used together to estimate the price that property would achieve, along with an interval expressing how certain that estimate is.

Proposed Machine Learning Task

The project uses a supervised regression task in order to achieve the business objective of producing trustworthy per-property price estimates.

Input: Property attributes location, lot, size, quality and condition, age, amenities, and any engineered features derived from these.

Output: For each property

1. A point estimate of sale price.
2. A prediction interval around that estimate, at a stated nominal level such as 80%.
3. A decomposition of the estimate into business-meaningful attribute groups.
4. A routing flag of AUTO or ESCALATE, derived from the interval width.

Point estimate and interval

The point estimate comes from a standard regression model. The interval is produced separately, by training quantile regression models at the lower and upper quantiles corresponding to the chosen nominal level. The lower and upper models give the interval endpoints for that specific property, which is why interval width varies from property to property rather than being a fixed band.

If time permits, a conformal calibration step will be applied on top of the quantile models to obtain a stronger coverage guarantee. Note that a naive conformal approach that assigns every property the same interval width would defeat the purpose of this project, since no property would ever be distinguishable as unusual; the nonformalized quantile approach preserves varying widths and is the one that will be used if this component is attempted.

Decomposition

The decomposition is produced using additive feature attributions, which by construction sum with a base value to recover the model's prediction exactly. The individual attributions are then grouped into the business-meaningful categories listed in the Dataset section, so that the agent sees a handful of interpretable quantities rather than dozens of raw feature contributions.

A land-versus-structure summary will also be reported, collapsing the groups into value attributable to location and lot versus value attributable to the building itself. This distinction is directly actionable: a property whose value is dominated by location has little renovation headroom, while a property whose value is dominated by the building has more.

Target transformation and its consequences

The sale price distribution is expected to be right-skewed, and a logarithmic transformation of the target is anticipated. This creates two technical consequences that will be handled explicitly rather than ignored:

- Point estimates. Exponentiating a prediction made on the log scale does not recover the mean price; it approximates the median. Either the estimate will be reported as a median, or a correction will be applied. This will be stated plainly in the report rather than left implicitly.
- Interval endpoints. Quantiles are preserved under monotone transformation, so interval endpoints computed on the log scale back-transform correctly without adjustment. This is a reason to be comfortable modelling on the log scale and will be noted as such.

Scope of committed output

Output component	Status
Point estimate	Core
Prediction interval	Core
Price decomposition	Core
Comparable sales retrieval	Stretch — only if core is complete and evaluated by end of Week 4
Improvement lever ranking	Stretch — same condition

The two stretch components are scoped separately because the descriptor requires the project to remain manageable within seven weeks. The group considers it better to scope honestly now than to over-promise in this submission and under-deliver in the final report.

Proposed Model Strategy

A baseline and at least three alternatives will be trained and compared, as required by the descriptor. Each model is included for a stated reason rather than lengthening the comparison.

Candidate models

Model	Role in the comparison
Dummy regressor (predicts the mean or median)	Trivial benchmark. Establishes the floor that every other model must clear
Ordinary least squares on a small set of obvious predictors	Naive but honest baseline. If a complex model cannot clearly beat this, that is itself a finding
Regularized linear regression (Ridge, Lasso, ElasticNet)	Handles the wide encoded feature matrix, resists overfitting, and provides directly interpretable coefficients for the valuation team
Random Forest regressor	Captures interactions between attributes without manual feature crossing; robust to outliers and to skewed inputs

Gradient boosting regressor	Typically, the strongest performer on tabular data of this size and structure
Quantile gradient boosting (lower and upper quantiles)	Produces the per-property prediction intervals that the routing decision depends on. Without this component the primary lens cannot be delivered

The same preprocessing pipeline and the same cross-validation folds will be used for every model, so that differences in reported performance reflect the models rather than differences in data handling.

Evaluation metrics

Accuracy alone is not a sufficient basis for choosing a model in this project, because the project's value depends on the reliability of the uncertainty estimates as much as on the accuracy of the point estimates.

Point accuracy:

- RMSE on the transformed target, comparable with published work on this dataset.
- MAE in currency, directly interpretable by the valuation team, tells them the typical error in money.
- MAPE expresses error proportionally, which matters because an error of a given size means something different on a low-value property than on a high-value one.

Interval quality:

- Empirical coverage, the proportion of properties whose actual sale price falls inside the nominal interval. If the nominal level is 80% and coverage is materially different, the intervals are miscalibrated.
- Mean interval width, reported alongside coverage. Coverage alone can be trivially satisfied by making every interval extremely wide, which would be useless; both must be reported together.

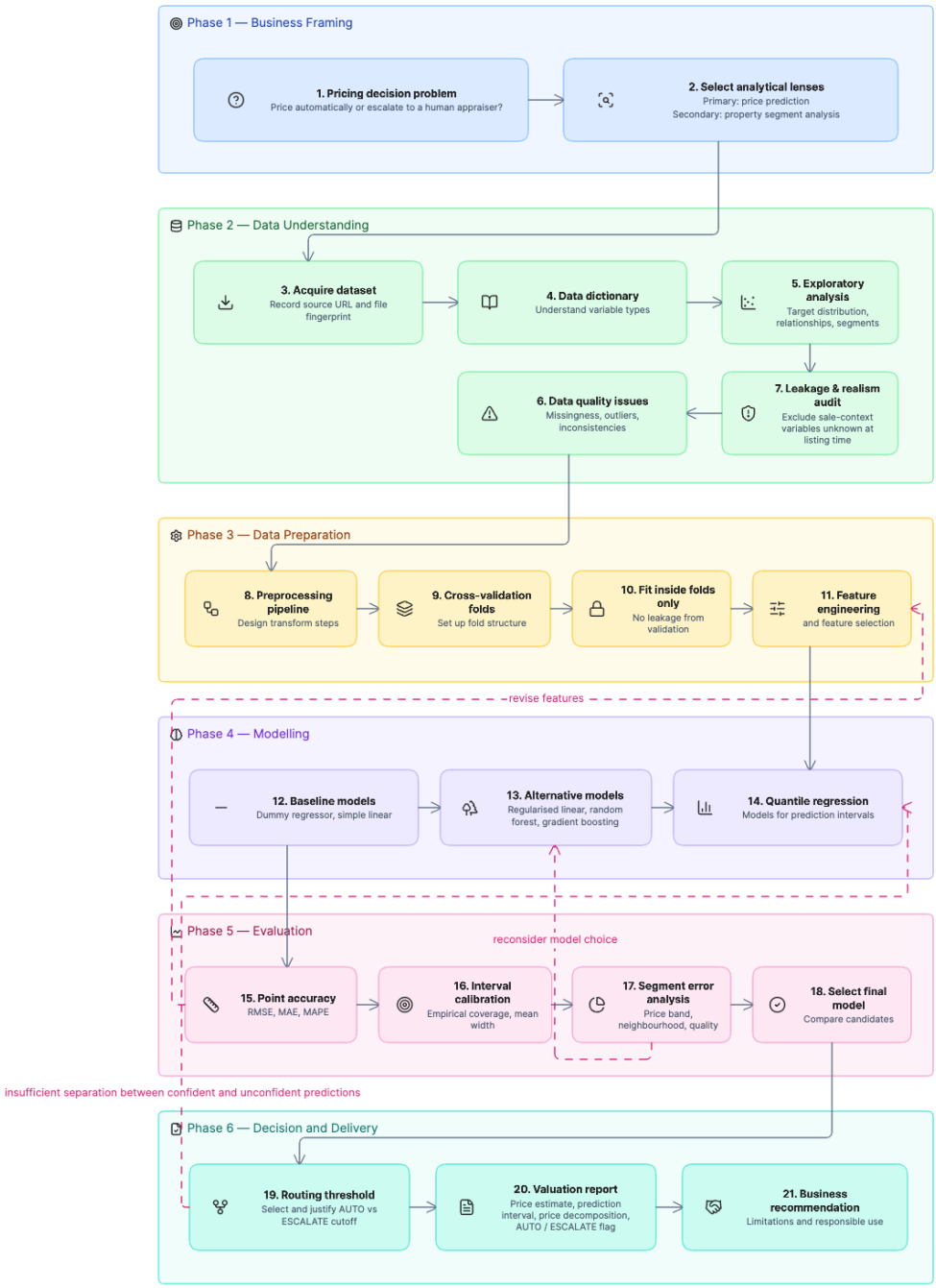
Segment behavior:

- Error broken down by price band, neighborhood, and quality level, an aggregate score can conceal complete failure on a subgroup. This is where the secondary lens contributes.

Decision value:

- Share properties routed to AUTO, and the error realized on that share, this is the quantity the stakeholder ultimately cares about, and it is the basis on which the routing threshold will be selected.

Project Workflow



Validation design

The models will be compared using repeated k-fold cross-validation on the labelled data, stratified by price band so that each fold contains a comparable mix of low, mid, and high-value properties.

The entire preprocessing pipeline will be fitted inside each training fold only. Any transformation fitted on the full dataset before splitting — scaling parameters, imputation values, or category encodings derived from the target — would leak information from the validation fold into training and would inflate the reported results. Ensuring this is a named responsibility in the group, and it is checked by a member other than the one who built the pipeline.

A secondary temporal validation will also be run, training on earlier sale years and testing on later ones, to assess whether the model's performance survives a shift in market conditions. The dataset's sale period is expected to span a period of market disruption, which makes this test meaningful. If the two validation designs disagree, that disagreement will be reported as a finding rather than resolved in favor of the more flattering result.

Hyperparameter selection will be carried out within the cross-validation structure and not on the same folds used to report results.

Final Evaluation

After the final model has been selected using cross-validation, its performance will be reported on the same cross-validated basis, since no labelled holdout set is available from the supplied test file.

The final evaluation will report:

- Point accuracy in both transformed and currency terms.
- Interval coverage against the nominal level, and mean interval width.
- Error broken down by property segment.
- The selected routing threshold, the share of the portfolio it routes to AUTO, and the pricing error realized on that share.
- A worked demonstration of the full valuation report output on three or four representative properties, chosen to include at least one that the model routes to ESCALATE.

The project concludes with interpretation of the results, a business recommendation, and a discussion of limitations and responsible use. The project ends at the point of producing decision support. No pricing system is deployed and no actual property is priced as part of this project.

Core Responsibilities of Group Members

Responsibilities are divided among the group members according to the stages of the machine learning process.

Member	Core Responsibility
IT24103124	Business problem framing, lens selection and justification, stakeholder analysis, exploratory data analysis
IT24102584	Data understanding, data quality assessment, preprocessing pipeline, feature engineering, leakage and realism audit
IT24102616	Model development, training, quantile interval models, hyperparameter tuning, model comparison
IT24103989	Evaluation, interval calibration analysis, segment error analysis, routing threshold selection, recommendation, limitations, responsible AI

Although each member has a primary area, all members contribute to the project and major decisions are discussed by the group so that the business problem, data preparation, modelling, evaluation, and recommendation remain consistent with one another.

Separation of duties. The member who builds the preprocessing pipeline is deliberately not the member who runs the evaluation. Leakage errors are hardest to see for the person who wrote the code, so the evaluation owner independently verifies that no transformation is fitted outside the cross-validation folds.

IT24103124 — Business Problem and Exploratory Analysis

- Defining the business problem: establishing the pricing decision the project supports and the cost of getting it wrong in each direction.
- Justifying the lens selection: documenting why price prediction was chosen over the alternative lenses, and why property segment analysis is adopted in a supporting role rather than as a parallel study.
- Identifying stakeholders: the valuation team, listing agents, and appraisers, and what each of them needs from the output.
- Defining the unit of analysis: confirming that each observation is one property sale.
- Conducting exploratory data analysis: examining distributions, relationships, and patterns in the property data.
- Investigating the target distribution: assessing the skewness of sale price and the case for a transformation.
- Investigating relationships with sale price: examining how location, size, quality, age, and amenity attributes relate to price.
- Investigating segment structure: identifying property categories that may behave differently, which supports the secondary lens.
- Maintaining the EDA insight log: recording each finding with its evidence and its implication for the task.

IT24102584 — Data Preparation

- Understanding the dataset structure: rows, columns, variable types, and the meaning of one observation.
- Building the data dictionary: documenting every variable with its meaning, type, and units.
- Identifying variable types: distinguishing numerical, nominal, ordinal, and binary variables and determining appropriate treatment for each.
- Separating absent features from missing values: determining, for each variable using a missing marker, whether it indicates absence of a feature or genuinely unrecorded data and treating each case appropriately.
- Handling genuinely missing values: selecting and justifying an imputation approach, compared against a simpler alternative.

- Encoding categorical variables: ordered encoding for genuine ordinal scales, and an appropriate approach for high-cardinality nominal variables.
- Conducting feature engineering: creating derived features such as total area, total bathroom count, and property age, each with a stated reason.
- Auditing for leakage and realism: verifying that no variable derived from the sale price enters the predictors, and assessing which variables would genuinely be known at the time a listing price is set.
- Building the preprocessing pipeline: implementing all transformations as a single pipeline object so that everything is fitted inside training folds only.
- Maintaining the preprocessing and feature log: recording each decision with the options considered, the choice made, and the evidence.

IT24102616 — Model Development

- Selecting candidate algorithms: choosing models that differ meaningfully in their assumptions, interpretability, and capacity, rather than assembling a long list.
- Training the baseline models: establishing the trivial and naive benchmarks that every subsequent model must beat.
- Training the alternative models: fitting the regularized linear, tree ensemble, and boosting models under a common pipeline and common folds.
- Training the quantile models: fitting the lower and upper quantile regressors that produce the per-property prediction intervals.
- Applying cross-validation: running the repeated stratified cross-validation used for all comparisons.
- Hyperparameter tuning: Search within the cross-validation structure, not on the folds used for final reporting.
- Producing the price decomposition: computing additive feature attributions and grouping them into the business categories, including the land-versus-structure summary.
- Investigating feature importance: examining which attributes drive predictions, where the model permits it.
- Documenting the modelling process: recording algorithms, settings, tuning decisions, and the reasoning behind each.

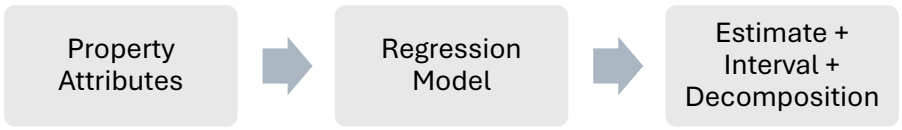
IT24103989 — Evaluation and Business Interpretation

- Evaluating point accuracy: reporting RMSE, MAE in currency, and MAPE across cross-validation folds with dispersion, never from a single split.
- Evaluating interval calibration: measuring empirical coverage against the nominal level and reporting it alongside mean interval width.
- Analyzing segment error: breaking error down by price band, neighborhood, and quality level to identify where the model fails.
- Running temporal validation: testing whether performance survives a shift in market period, and reporting disagreement with the primary design as a finding.
- Selecting the routing threshold: examining the trade-off between the share of properties auto priced and the error realized on that share and justifying the chosen operating point.
- Verifying leakage prevention independently: checking that the preprocessing pipeline is fitted inside folds only, as a check on the pipeline owner's work.
- Selecting the final model: supporting the decision using validation results and the project's business requirements, not accuracy alone.
- Interpreting results: explaining what the numbers mean for the valuation team.
- Developing the recommendation: translating findings into a concrete operating recommendation with its expected benefit and its risks.
- Identifying limitations: documenting limitations of the dataset, the model, the uncertainty estimates, and the decomposition.
- Responsible AI: ensuring the output is positioned as decision support, and that the limits of the decomposition as an explanatory tool are stated accurately.

Expected Project Outcome

The expected outcome is a machine learning approach that estimates residential property sale prices and reports, for each estimate, how much confidence should be placed in it and what is driving it.

The target-generation process is direct: the recorded sale price is the target, and no derived labelling step is required.



Property Attributes → Regression Model → Estimate + Interval + Decomposition

Sale-context variables are excluded from the predictors because they are not available at the time a listing price must be set. Including them would raise measured accuracy while producing a model that could not be used for its stated purpose.

The expected outcome addresses the business problem directly. The valuation team receives, for each property, a price estimate they can act on, an honest signal of when that estimate should not be trusted, and a breakdown that makes the estimate reviewable by a person.

The recommendation arising from the project will take the form of an operating rule: route a stated share of the portfolio to automatic pricing, keep the remainder with human appraisers, and expect a stated level of pricing error on the automated share.

It is essential that the output is used as decision support rather than as a decision-maker. A property should not be priced solely on a model output without human review, and the routing rule exists precisely to identify the cases where human review is most needed.

Initial Limitations

Several limitations apply to this project, particularly at this initial stage.

1. Dataset Generalizability

The dataset records sales in a single city over a limited period. A model trained on it reflects that market's conditions, price levels, and property mix. It should not be assumed to transfer to other cities or to the present day without revalidation.

2. Market Period

The sale period covered by the dataset is expected to span a period of significant housing market disruption. This makes the temporal validation described earlier informative, but it also means that overall price levels learned by the model are tied to that period and would require recalibration for current use.

3. Absence of Market Context Variables

The dataset describes properties but records no information about market conditions at the time of sale — interest rates, local inventory, time on market, or the number of competing listings. Two identical properties sold in different market conditions would achieve different prices, and the model cannot account for this. Some of this variation will surface as apparently irreducible uncertainty in the prediction intervals.

4. Dependence on Interval Calibration

The routing decision depends entirely on the prediction intervals being well calibrated. If the intervals are systematically too narrow, properties will be routed to automatic pricing that should have been escalated, and the agency will be given false confidence. Empirical coverage will therefore be measured and reported rather than assumed, and the failure mode will be stated explicitly in the final report.

5. Limits of the Price Decomposition

The decomposition shows how the model arrives at its prediction. It does not establish why a property is worth what it is worth in the real world. Where property attributes are correlated with one another — and several quality-related attributes are expected to be — the attribution of value between them may be arbitrary in ways that look meaningful. The decomposition will therefore be presented as a description of model behavior, not as a causal account of property value.

6. Association Rather Than Causation

If the improvement-lever component is attempted, its outputs describe associations observed in past sales, not the effect of carrying out a renovation. The dataset contains no renovation costs

and no before-and-after observations of the same property, so no return on investment can be calculated. This will be stated in the wording of any such output, not only in a footnote.

7. Prediction Error

The model will make errors in both directions. Overestimating leads to properties that do not sell; underestimating loses money for the seller. Neither error is negligible, and the routing rule is the project's mechanism for limiting exposure to both on the properties where the model is least reliable.

8. Responsible Use

The model output supports a pricing decision; it does not make one. Property valuation affects individuals' finances significantly, and an automated estimate should be reviewable, explainable, and overridable by a human. The decomposition

Conclusion

This project proposes a machine learning approach to property valuation within the Real Estate domain, using the Ames Housing dataset supplied for group code 8.

The primary business lens is price prediction, selected because it is the only one of the available lenses that produces a per-property output the valuation team acts on. The secondary lens, property segment analysis, is adopted solely to identify which categories of property the model cannot price confidently, which is what the routing decision requires.

The project's distinguishing feature is not the prediction itself, but what accompanies it. Each estimate is delivered with a prediction interval expressing how certain the model is about that specific property, and a decomposition showing which groups of attributes the value comes from. The interval drives a concrete operating decision: price the property automatically or send it to a human appraiser.

The machine learning task is supervised regression on the recorded sale price. A trivial benchmark and a naive baseline establish the floor, and regularized linear, tree ensemble, and boosting models are compared against them under a common preprocessing pipeline and common cross-validation folds. Quantile regression models produce prediction intervals.

Evaluation goes beyond point accuracy. Alongside RMSE, MAE in currency, and MAPE, the project measures whether the prediction intervals are honest — reporting empirical coverage against the nominal level together with mean interval width, since either figure alone can be made to look good at the expense of the other. Error is broken down by property segment, and a secondary temporal validation tests whether performance survives a shift in market conditions.

The supplied test file is unlabeled, so all reported performance comes from cross-validation on the labelled data. The entire preprocessing pipeline is fitted inside training folds only, and this is verified by a member other than the one who built it.

The final objective is a data-driven approach to pricing that tells the valuation team not only what a property is likely to be worth, but how far that figure can be trusted and what is driving it. The output supports the pricing decision; it does not replace the person making it.